Recurring Course Themes

Indexing vs. Search

Indexing allows finding data more quickly.

But requires time to establish & maintain.

More- vs. Less-Structured Design

Design can precisely model current semantics.

- More efficient data access.
- Enforce data integrity.

Costs:

- Requires time.
- Design complexity.
- Less flexible to changes.

Data Integrity vs. Flexibility

Can enforce constraints on data.

- More efficient data use.

Costs:

- Design complexity.
- Less flexible to changes.

Themes Interrelated

E.g., emphasizing data integrity typically leads to a more structured design and using more indexing.